

Linear Predictive Coding (LPC)

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Introduction to LPC

Definition

Linear Predictive Coding (LPC) is one of the most important techniques in speech processing for modeling and representing speech signals.

Main applications:

- Speech coding and compression
- Formant estimation
- Pitch analysis
- Speech synthesis
- Speaker recognition
- Vocoders

Motivation for LPC

Speech signals contain a large amount of redundancy.

Main Idea

A speech sample can be approximated using a weighted combination of previous speech samples.

Advantages:

- Compact representation
- Efficient coding
- Separation of source and filter
- Spectral envelope estimation

Source-Filter Representation

Speech can be represented as:

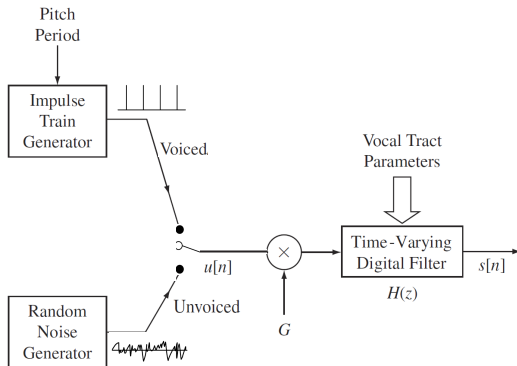
$$S(z) = H(z)U(z)$$

where:

$U(z)$: excitation signal

$H(z)$: vocal tract transfer
function

$S(z)$: speech signal



Block diagram of simplified model for speech production

Interpretation

The excitation source is shaped by the vocal tract resonances to produce speech.

All-Pole Vocal Tract Model

The vocal tract (VT) is modeled using an all-pole filter:

$$H(z) = \frac{G}{1 - \sum_{k=1}^p a_k z^{-k}}$$

or:

$$H(z) = \frac{G}{A(z)}$$

with:

$$A(z) = 1 - \sum_{k=1}^p a_k z^{-k}$$

a_k : VT system coefficients

p : VT model order

G : gain factor

Linear Prediction Principle

The current sample $s[n]$ is predicted from past samples:

$$\hat{s}[n] = \sum_{k=1}^p a_k s[n - k]$$

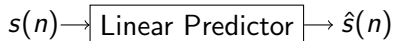
where:

$\hat{s}[n]$: predicted sample

$s[n - k]$: previous speech samples

a_k : predictor coefficients

p : prediction order



Objective

Determine the coefficients that minimize the prediction error.

Prediction Error Signal and its characteristics

The prediction error is: $e[n] = s[n] - \hat{s}[n]$

thus:
$$e[n] = s[n] - \sum_{k=1}^p a_k s[n - k]$$

Interpretation

The residual signal contains the information that cannot be predicted from previous samples.

For voiced speech:

- Residual becomes impulse-like
- Pitch information is preserved

For unvoiced speech:

- Residual resembles white noise

Important

The residual approximates the excitation signal.

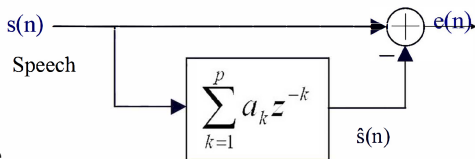
Inverse Filter Interpretation and Role

Since: $H(z) = \frac{G}{A(z)}$ then: $S(z) = \frac{G}{A(z)} U(z)$

Applying the inverse filter: $E(z) = A(z)S(z)$ gives: $E(z) \approx GU(z)$

The inverse filter:

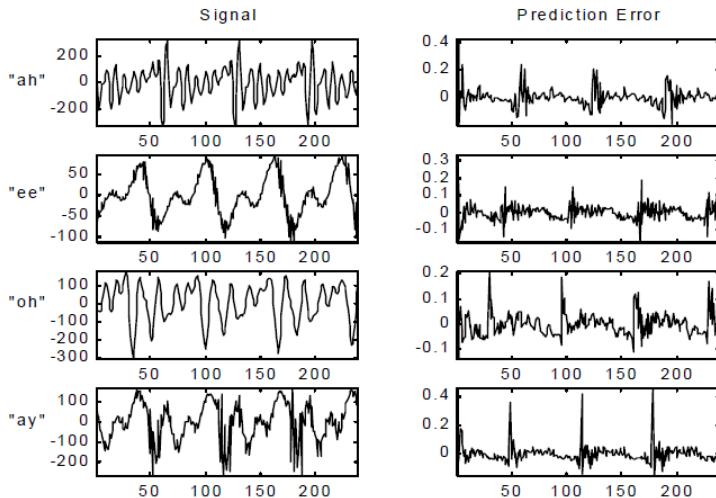
- Removes vocal tract resonances
- Suppresses the spectral envelope
- Extracts the excitation signal. For voiced speech, $e[n]$ has large peaks at pitch period boundaries. For unvoiced speech, $e[n]$ scaled random noise.



Physical Meaning

The vocal tract effect is cancelled, leaving mainly pitch pulses or noise excitation. Used in analysis-by-synthesis coders such as Code-Excited Linear Prediction (CELP).

Inverse Filter Interpretation and Role (continued)



LPC prediction error signals $e[n]$ for several vowels.

LPC Spectrum Formant Estimation

The LPC spectrum is obtained from: $|H(e^{j\omega})|$

Characteristics:

- Smooth spectral envelope
- Harmonic structure is suppressed
- Formants become visible

Formants correspond to resonances of the vocal tract.

Key Property

The poles of the LPC filter determine the formant frequencies.

Pole characteristics:

- Pole angle $\rightarrow$ formant frequency
- Pole radius $\rightarrow$ formant bandwidth

Mean Squared Error and Derivation of LPC Equations

The LPC coefficients are obtained by minimizing: $E = \sum_{n=0}^{N-1} e^2[n]$

where:
$$e[n] = s[n] - \sum_{k=1}^p a_k s[n-k]$$

Optimization

Find the coefficients a_k that minimize the prediction error energy.

The optimal coefficients satisfy: $\frac{\partial E}{\partial a_i} = 0$

This leads to:
$$\sum_{n=0}^{N-1} e[n]s[n-i] = 0 \quad \text{for: } i = 1, 2, \dots, p$$

Orthogonality Principle

The prediction error is orthogonal to previous speech samples.

Autocorrelation Function

Autocorrelation is defined as:

$$R(i) = \sum_{n=0}^{N-1-i} s[n]s[n+i]$$

where $R(i)$, i and N are the autocorrelation coefficient, the lag and the frame length, respectively.

Main properties:

- Symmetric: $R(i) = R(-i)$
- Maximum at zero lag $R(0) \geq R(i)$
- Periodicity appears for voiced speech

Using autocorrelation, we obtain:

$$\sum_{k=1}^p a_k R(i-k) = R(i) \quad \text{for } i = 1, 2, \dots, p$$

In LPC, these equations are known as Yule-Walker equations

Matrix Form of LPC Equations

The LPC equations can be written as: $\mathbf{R}\mathbf{a} = \mathbf{r}$

where: $\mathbf{R} = \begin{bmatrix} R(0) & R(1) & \cdots & R(p-1) \\ R(1) & R(0) & \cdots & R(p-2) \\ \vdots & \vdots & \ddots & \vdots \\ R(p-1) & R(p-2) & \cdots & R(0) \end{bmatrix}$

LPC coefficient vector: $\mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_p \end{bmatrix}$ Autocorrelation vector: $\mathbf{r} = \begin{bmatrix} R(1) \\ R(2) \\ \vdots \\ R(p) \end{bmatrix}$

$\mathbf{R}$: autocorrelation matrix; $\mathbf{a}$: LPC coefficient vector; $\mathbf{r}$: autocorrelation vector

Toeplitz Structure of R and Need for Recursive Algorithms

The autocorrelation matrix has a Toeplitz structure:

$$R(i, j) = R(|i - j|)$$

Properties:

- Symmetric matrix
- Constant diagonals
- Positive definite

Advantage

Efficient recursive algorithms can be used.

Direct inversion of R : $\mathbf{a} = \mathbf{R}^{-1}\mathbf{r}$ is computationally expensive.

Problem: The complexity increases rapidly with prediction order p

Solution: Levinson-Durbin recursion

Levinson-Durbin Algorithm

The algorithm recursively computes:

- LPC coefficients
- Reflection coefficients
- Prediction error energy

Main quantities:

$a_k^{(i)}$: LPC coefficient at iteration i

k_i : reflection coefficient

E_i : prediction error energy

Initialization Step

Initial conditions: $E_0 = R(0)$

Initial LPC coefficients: $a_0^{(0)} = 1$

Interpretation

The initial prediction error equals the signal energy.

Levinson-Durbin Algorithm (continued)

At iteration i :

$$k_i = \frac{R(i) - \sum_{j=1}^{i-1} a_j^{(i-1)} R(i-j)}{E_{i-1}}$$

where k_i : reflection coefficient (PARCOR coefficients) and E_{i-1} : previous prediction error. The LPC coefficients are updated recursively:

$$a_i^{(i)} = k_i \quad \text{and} \quad a_j^{(i)} = a_j^{(i-1)} - k_i a_{i-j}^{(i-1)} \quad \text{for: } j = 1, 2, \dots, i-1$$

Prediction error energy is updated as: $E_i = (1 - k_i^2) E_{i-1}$

Normalized error: $V^{(i)} = E^{(i)} / R^{(0)}$.

Stability condition

The filter is stable if and only if $|k_i| < 1$ for all i ($i = 1, \dots, p$).

Advantages of Levinson-Durbin algorithm

- Low computational complexity $\mathcal{O}(p^3) \rightarrow \mathcal{O}(p^2)$
- Exploits Toeplitz structure : Avoids expensive full matrix inversion
- Filter stability:
 - Reflection coefficients satisfy: $|k_i| < 1$
 - Ensures LPC poles remain inside the unit circle
- Fast and memory efficient
 - Highly amenable to hardware implementation
 - Suitable for real-time processing
- Provides reflection (PARCOR) coefficients
 - Useful for lattice filters and speech coding
- Useful for LPC gain estimation and residual analysis

Reflection Coefficients and Vocal Tract

Reflection coefficients model reflections inside the vocal tract.

$$r_i = \frac{A_{i+1} - A_i}{A_{i+1} + A_i}$$

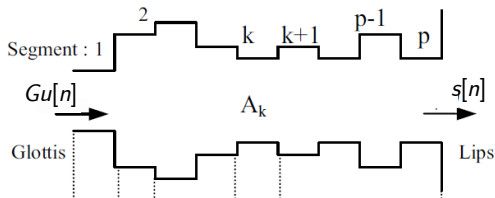
where: A_i : tube cross-sectional area

r_i : reflection coefficient (measures energy reflected at each junction.)

Related to Durbin's k_i : $r_i = -k_i$.

- If $A_{i+1} = A_i$ then $r_i = 0$ meaning no reflection occurs.
- Large area changes produce stronger acoustic reflections.

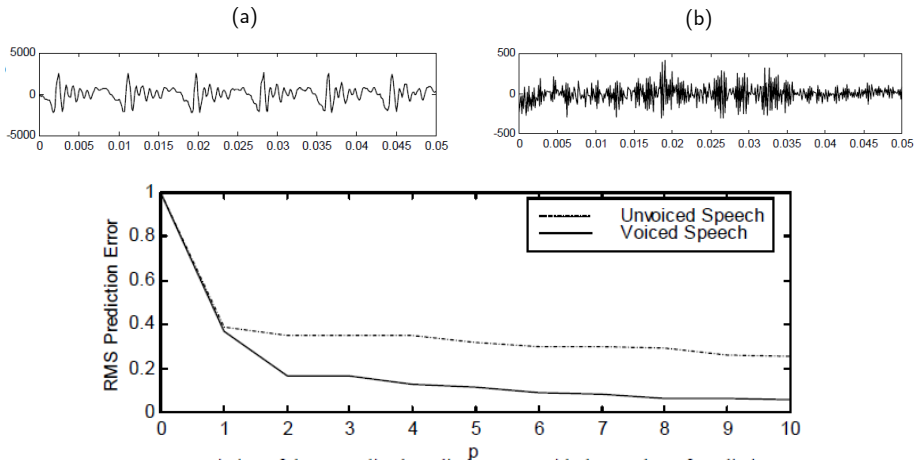
Human vocal tract can be approximated as lossless tubes of varying cross sectional areas A_i



Typical LPC analysis stages:

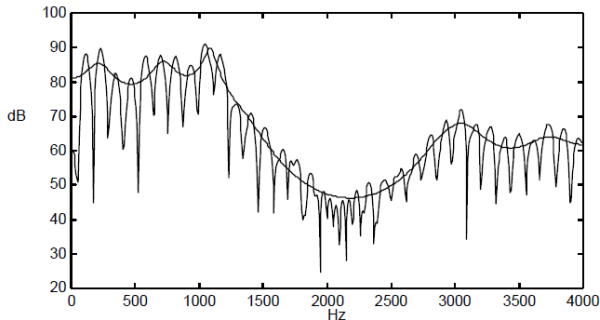
- Pre-emphasis
- Framing
- Windowing
- Autocorrelation computation
- Levinson-Durbin recursion
- LPC coefficient estimation

LPC : Results illustration



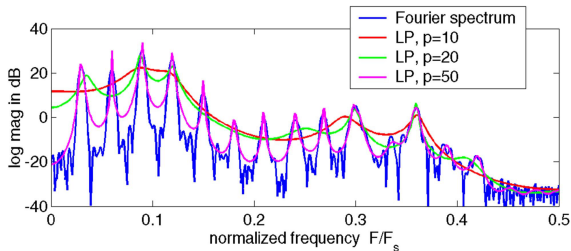
Variation of the normalized prediction error with the prediction coefficients number p for voiced and unvoiced speech segments (a) and (b). Auto-correlation method was used with a 30 ms Hamming window and a sampling rate of 8 kHz. The interpretation of this results is that unvoiced speech is less linearly predictable than voiced speech.

LPC : Results illustration



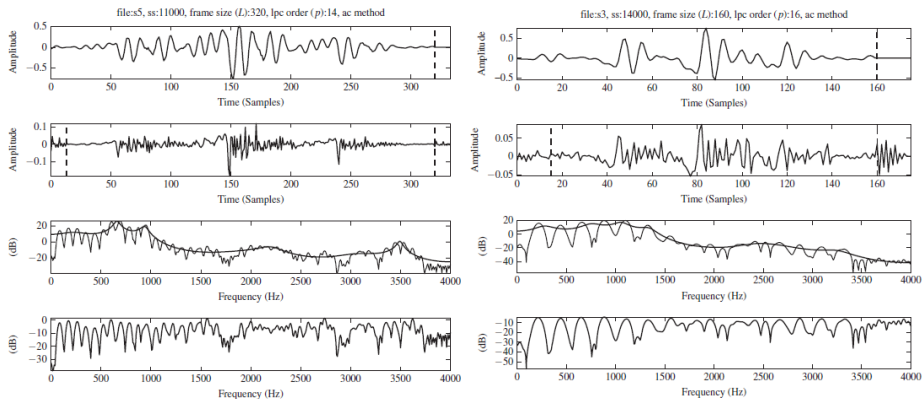
LPC spectrum of the /ah/ phoneme. Here are used 30-ms Hamming window and the autocorrelation method with $p=14$.

The short-time spectrum is also shown.



LP Analysis is seen to be a method of short-time spectrum estimation with removal of excitation fine structure (a form of wideband spectrum analysis)

LPC : Results illustration



Typical signals and spectra for the use of the linear predictive autocorrelation methods for a male (left figure) and female (right figure) speakers. The upper panels show the speech frame being analyzed; the second panels show the prediction error resulting from a $p = 14$ (for male), $p = 16$ (for female) linear prediction analysis; the third panels show the log magnitude spectrum of the signal along with the linear prediction model spectrum; the bottom panels show the log magnitude spectrum of the error signal.

Applications of LPC

LPC is used in:

- Speech coding
- Vocoders
- Formant tracking
- Pitch estimation
- Speaker recognition
- Speech synthesis
- Mobile communications

Advantages and Limitations

Advantages:

- Compact representation
- Efficient computation
- Good spectral modeling

Limitations:

- All-pole assumption
- Sensitive to noise
- Less accurate for nasal sounds

- LPC models speech using linear prediction
- LPC coefficients represent the vocal tract
- The residual approximates the excitation
- Levinson-Durbin efficiently computes the coefficients
- LPC remains fundamental in speech processing

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